Istituto di Ricerca e Certificazione per le Costruzioni Sostenibili
Notified Body No. 1994 in accordance with Regulation CPR (EU) No. 305/2011

TEST REPORT

Number:

1994-CPR-RP2157

Issuing date:

13 January 2021

Applicant:

EVEREST SHPK
BULEVARDI BLU, 544
1030 KAMZA, TIRANA, ALBANIA

Tested product:

Double-casement sliding door
with Trade Name
"PB2AAR_EVEREST_91000"
(cf. description)

Executed tests:

Air permeability
Watertightness
Resistance to wind load

Normative references:

EN 14351-1:2006+A2:2016
EN 1026:2016 **EN 12207:2017**
EN 1027:2016 **EN12208:1999**
EN 12211:2016 **EN 12210:2016**

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The results included in the Test Report refer exclusively to the tested objects, identified in the present Test Report.
Translation issued on 2021-04-06

1 Description of the tested sample

The test specimen is a Double-casement sliding door with Trade Name “PB2AAR_EVEREST_91000”.
The specimen was identified by the applicant in accordance with product standard EN 14351-1:2006+A2:2016.
The identification code of the test specimen, as declared by the applicant, is: “172020”.
Laboratory Code: “PF2ASA_EVEREST_91000”.

The description and the technical drawings that follow, referred to the test specimen, were declared/supplied by the applicant under his own responsibility:

- Material: aluminium-alloy profile EN-AW 6060-T5;
  Stato di fornitura T5 secondo UNI EN 12020-2:2002
  Profiles:

frame 2-way with collect condensate	item 91106
inserted-glass sash	item 91202
inserted-glass sash with reinforcement	item 91204
central labyrinth	item 91501
track cover in PVC	item EVE10617
central labyrinth in PVC	item EVE10717
rail cover	item EV504

  of applicant EVEREST, Albania.

- Corner joint: - alignment corner joint frame — item S060SS
  of applicant EVEREST, Albania.

- Glazings: double glazing with 4mm monolithic glass, 16 mm cavity and 4mm monolithic glass

- Glazing gaskets:

- internal glass gasket	item 1020
- external glass gasket	item L407EP

  of applicant EVEREST, Albania.

- Gaskets:

- Perimeter brush	item R126NB
- Central gaskets	item CP001368

  of applicant EVEREST, Albania.

- Water drainage system: cf. enclosed technical drawings.

- Hardware: locking system comprising no. 14 locking points:

- 2pz multipoint locking Fast Lock	item 6381
- 2pz locking kit 6381	item 6391A.20
- 4pz trolleys capacity 100kg per couple	item PP6620Z.96
- 1pz kit for assembling 2 movable sash	item PP6620.186
- 1pz Hammer	item D057AB
- 1pz handle fast lock	item 75201

  Supplied of EVEREST, Albania.

- Declares nominal dimensions: cf. enclosed technical drawings.

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INTERNAL VIEW

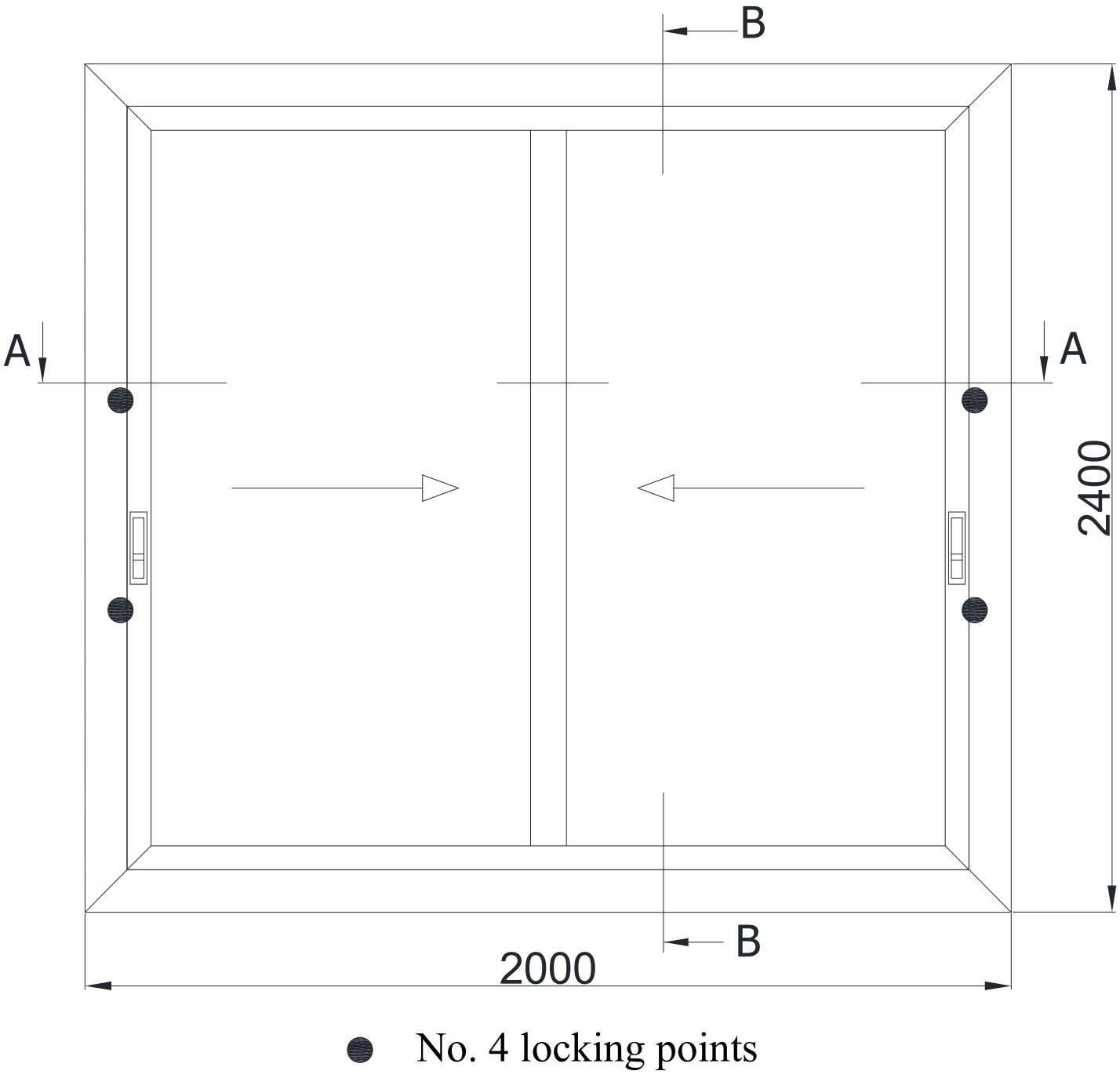


Fig. 1: Front view of the received and tested sample, positioning scheme of locking devices: front view (declared nominal dimensions, expressed in mm)

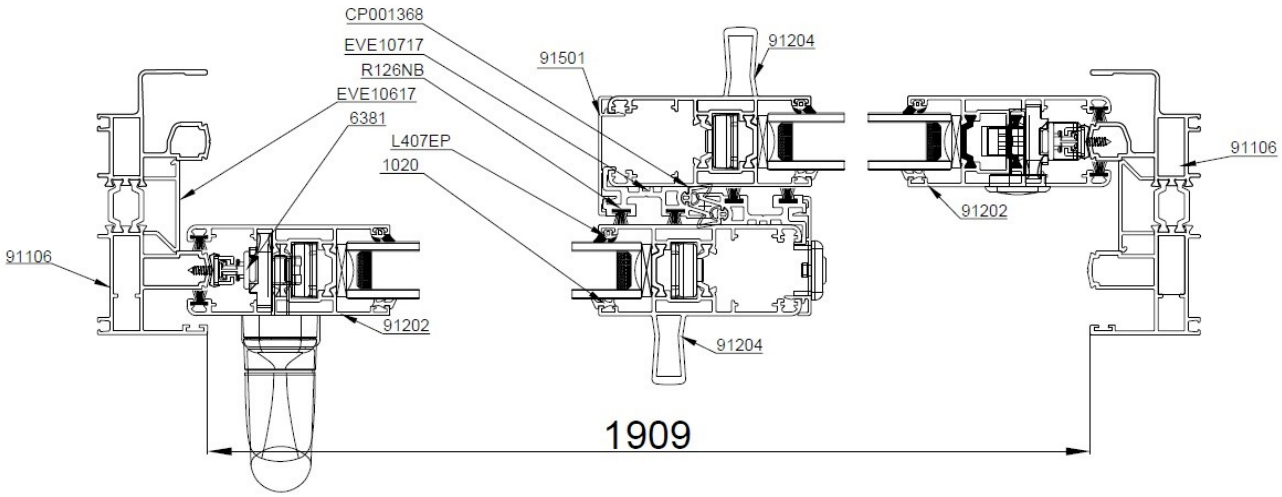


Fig. 2. Horizontal section AA of the tested sample
(declared nominal dimensions, expressed in mm)

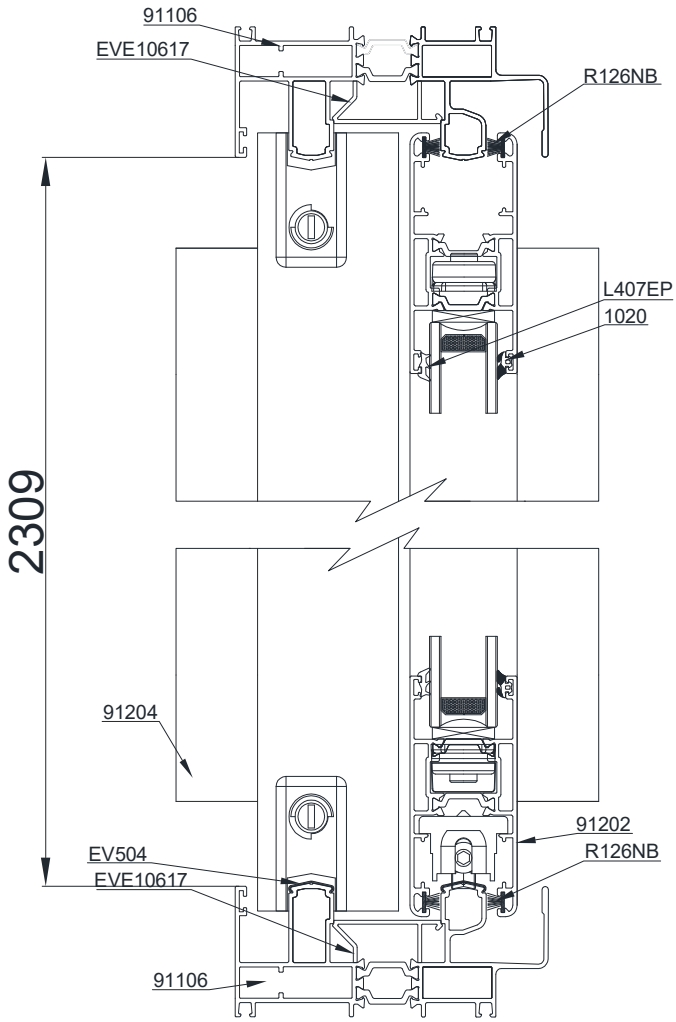


Fig. 3. Vertical section BB of the tested sample
(declared nominal dimensions, expressed in mm)

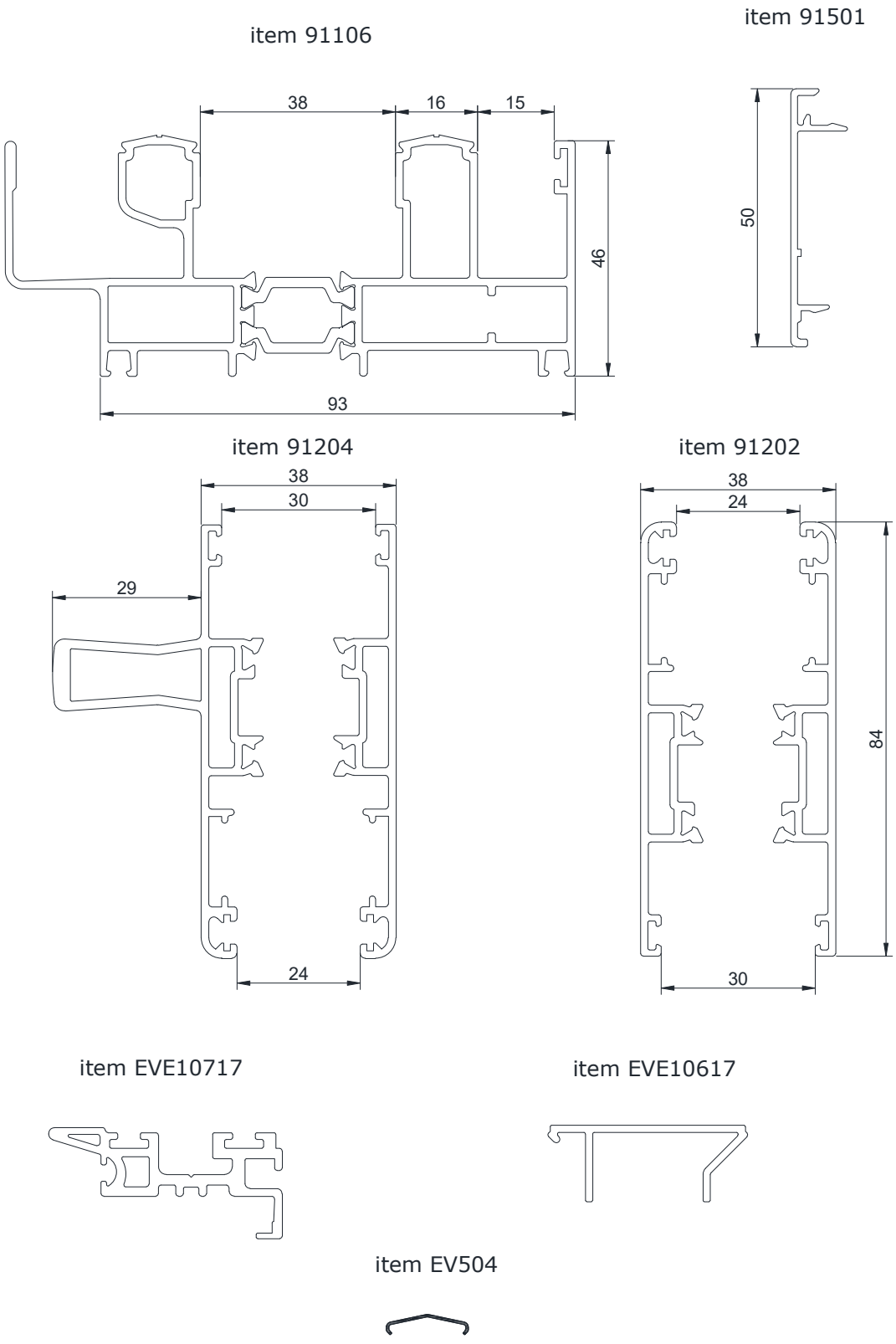


Fig.4: Graphical description of the profiles of the tested sample
(declared nominal dimensions, expressed in mm)

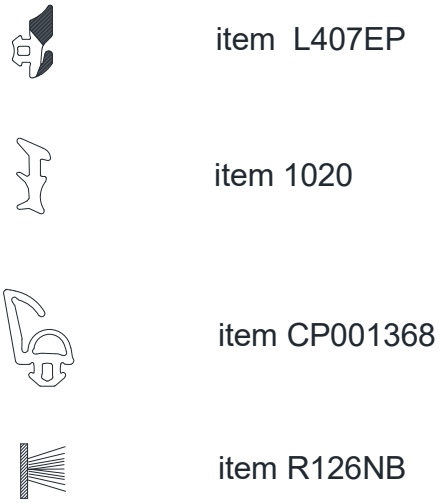


Fig. 4. Graphical description of the gaskets of the tested sample

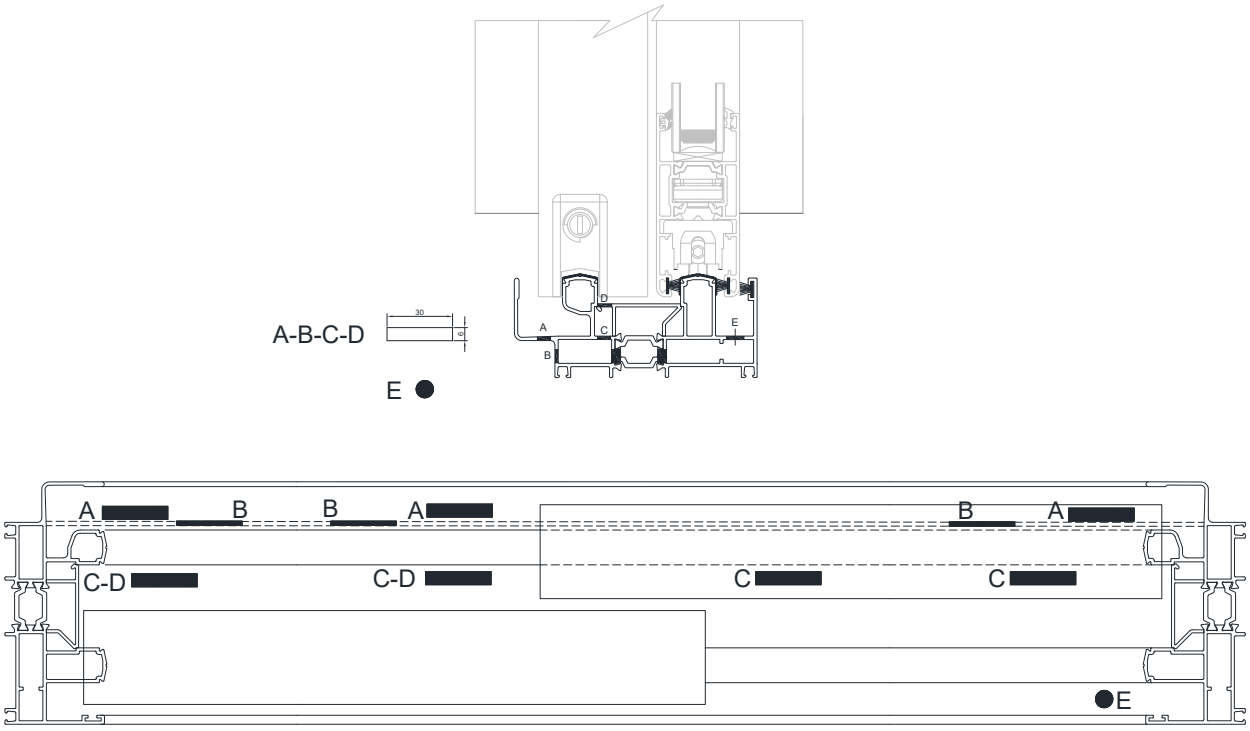


Fig. 5. Graphical description of the discharge slots of the tested sample
(declared nominal dimensions, expressed in mm)

2 Sampling procedure

The product was sampled directly by the applicant, who indicated its traceability on the basis of the previously described code.

3 Sample preparation procedure

The sample was prepared in accordance with the provisions of standards EN 1026:2016, EN 1027:2016 and EN 12211:2016. The sample was introduced by the applicant in a supporting frame stiff enough to withstand the test pressures, fastened as under service conditions and free from bending and torsion stresses that might affect test results. When delivered, the sample was stored in an appropriate area of the laboratory, then conditioned at controlled temperature and relative humidity within the limits provided for by the standards (between 10° C and 30° C and between 25% and 75% U.R. respectively), for more than 4 hours just before the test. The sample was then fastened to the testing equipment. A dimensional control of the sample was carried out prior to the test by means of a flexometer.

4 Test method

The performed and described tests were carried in reference to the article. 46 of the EU Regulation CPR n. 305/2011 in the laboratory of MasterLAB S.r.l., located in Conversano (BA), in 2020/11/27, directly by an operator of MasterLAB S.r.l., Antonio Delfine, in presence of Nicola Antonio Dentamaro, responsible for CE Marking, and under the instruction and with the supervision of the laboratory technician Stefano Galli of IRCCOS S.r.l. (NB1994), located in via Achille Grandi 17 a Samarate (VA).

4.1 Air permeability

The test was performed in accordance with standard EN 1026:2016, EN 12207:2017 (applying procedure IOP10 rev. 0).

- *Principle of test.* It consists in measuring the sample's air permeability subjected to an established sequence of pressures.
- *Test procedure.* In accordance with EN 14351-1:2006+A2:2016, two tests were carried out in sequence, one under positive pressure and one under negative pressure, in both cases according to the methods hereby described.
  The opening parts of the sample were opened and closed once, then blocked in the closed position. The test has developed according to two different phases, each one according to the same sequence, reported in the following: *the air permeability of the testing sample* (expressed in m^3/h) has been measured at different pressure levels, considering negligible values related to the measurement of air permeability of the testing chamber. In both cases, according to the testing sequence, three pressure pulses were applied with rising time not less than 1 second, each maintained for at least 3 seconds with value 10% more than the maximum test pressure; air permeability values were then measured and recorded under gradually increasing pressures with minimum intervals of 10 seconds, until the maximum pressure of 600 Pa, according to the following sequence: 50, 100, 150, 200, 250, 300, 450, 600 Pa. At each applied test pressure increment, the result of the measurements of air flow V_x was corrected on the basis of actual values of temperature T_x (expressed in °C) and atmospheric pressure P_x (expressed in kPa) measured during the execution of the test, in order to obtain air flow air flow (V_o) under normal conditions (T_o=293 K, P_o=101.3 kPa):

$$V_0 = V_x \times \frac{293}{273 + T_x} \times \frac{P_x}{101{,}3}$$

4.2 Watertightness

The test was performed in accordance with standard EN 1027:2016 and with reference to EN 12208:1999 (applying procedure IOP11 rev. 0).

- *Principle of test.* The test consists in the application of a constant and even amount of water on the external surface of the tested sample, while applying positive test pressure increments at pre-established

regular intervals (as defined below), during which pressure values, time and place of seepage are accurately recorded in order to determine the window's *watertightness limit*.

- *Test procedure.* The opening parts of the samples were opened and closed once, then blocked in the closed position. Three pressure pulses were applied with rising time not less than 1 second, each maintained for at least 3 seconds with value 10% more than the maximum test pressure and in any case not lower than 500 Pa. Water was projected by means of a row of nozzles with 400 mm ± 10 mm center-distance and average flow rate of 2 l/min each nozzle. The axis of the nozzles row was inclined as to the horizontal row by $(24_0^{+2})°$ in accordance with method 1A. Water was first delivered in the absence of pressure for 15 minutes, then the test pressure was applied by successive 5-minute steps, with increments of 50 Pa up to 300 Pa and of 300 Pa up with increments of 150 Pa until the sample's waterproofing limit was reached.

4.3 Resistance to wind load

The test was performed in accordance with standard EN 12211:2016 and with reference to EN 12210:2016 (applying procedure IOP12 rev. 0).

- *Principle of test.* The test consists in applying a defined series of positive and negative test pressures to measure and verify the relative frontal deformation and the resistance to damages caused by wind loads.
- *Test procedure.* The test was subdivided into three successive phases, illustrated below: *deformation test (under positive and negative pressure)*, *repeated pressure test* and *safety test*.

  - *Deformation test – Positive pressure*:
  Three air pressure pulses were applied with rising time not less than 1 second, each pulse maintained for at least 3 seconds and with a value 10% greater than the deformation pressure P1. After zeroing the instruments for frontal displacements, the sample was subjected to increasing test pressures with velocity not greater than 100 Pa/s in an incremental way up to pressure P1. Such pressure was maintained for 30 seconds during which the values of the frontal displacements of the characteristic points were measured and recorded. After taking the test pressure back to 0 Pa, with velocity not greater than 100 Pa/s and after (60 ± 5) s, the frontal residual deformations were measured and recorded.

  - *Deformation test – Negative pressure:*
  The sample was then subjected to negative test pressures decreasing down to pressure P1, following a procedure analogous to that adopted for the pressure test above.

  - *Repeated pressure test*:
  The sample was subjected to a series of 50 cycles including negative and positive pressures at value P2, according to the following sequence:
  - a first negative phase, followed by a positive phase, just like the last 50-pulse sequence;
  - the variation from – P2 to + P2 and vice versa was obtained in (7 ± 3)s;
  - value P2 was maintained for (7 ± 3)s.
  At the end of the 50 cycles, the movable parts of the sample were opened and closed to observe any damage or operational defect. Afterwards, the air permeability test was repeated in accordance with standard EN 1026:2016 by adopting the same procedure used for the previous test.

  - *Safety test*:
  The sample was subjected to a cycle of negative and positive test pressure under maximum pressure P3 according to the following sequence:
  - first, negative test pressure was applied;
  - the variation from 0 Pa to – P3 and vice versa was obtained in (7 ±3)s, maximum test pressure P3 was maintained for (7 ± 3)s;
  - the positive test pressure was applied after an interval of (7 ± 3)s with the same sequence.

5 Test equipment

The equipment used to perform the tests, in accordance with EN 1026:2016, EN 1027:2016 and EN 12211:2016, consists of:

- a wall with an open side to house the test sample;
- a device allowing to create a controlled pressure difference between the sample's faces;
- a device allowing to obtain a quick and controlled variation of pressure difference within specified limits;
- an instrument to measure the incoming and outgoing air-flow through the airtight chamber ;
- an instrument to measure pressure difference between the sample's faces;
- an instrument that measures the temperature inside the airtight chamber;
- an instrument that measures the temperature and relative humidity of the environment;
- an instrument that measures the atmospheric pressure of the environment;
- a device projecting water and allowing to create a continuous film of water all over the test surface by means of circular full-cone nozzles with the following features: $(120^{0}_{-10})°$ delivery angle and a 2 liter min/m^2 water flow rate;
- an instrument to control the amount of projected water;
- an instrument that measures water temperature;
- instruments that measure displacements;
- a device allowing to fix the measuring instruments and to ensure their stability during the test;

6 Expression of results

6.1 Air permeability

In accordance with 14351-1:2006+A2:2016, the results obtained are expressed as each test pressure as the arithmetic average of the values obtained in the two tests of air permeability at positive and negative pressure. In accordance with § 4 of the standard EN 12207:2017 and with reference to EN 1026:2016, for the classification of the sample the following criteria were respected:

- air permeability corrected as a function of the actual values of temperature and atmospheric pressure, was related both to the overall surface area of the sample (expressed in m^3/m^{2}h) and to the unit length of the opening joint (expressed in m^3/mh) and the arithmetic average of the values obtained in the two tests of air permeability at positive and negative pressure was therefore shown on a graph for each increase of test pressure;
- relevant classes were defined in accordance with the following Table, based on the100Pa reference pressure, where air permeability Q admitted at different test pressures P is determined using the following formula (where Q_{100} is the reference air permeability):

$$Q = Q_{100} \times \left(\frac{P}{100}\right)^{2/3}$$

- on the basis of test results, the window sample was assigned the relevant class when the obtained air permeability value did not exceed the upper limit fixed for that class at all intermediate test pressure levels up to maximum value as a function of the fulfilment of one of the following relations for the two curves, as reported in the bi-logarithmic diagram:
  - same class: the sample is classified in that class;
  - 2 contiguous classes: the sample is classified in the most favourable class;
  - 2-class difference: the sample is classified in the medium class;
  - more than 2-class difference: the sample shall not be classified.

Class	Maximum test pressure (Pa)	Reference air permeability 100 Pa (m^3/hm^2)	Reference air permeability 100 Pa (m^3/hm)
0	Not subjected to test		
1	150	50	12.50
2	300	27	6.75
3	600	9	2.25
4	600	3	0.75

Tab. 1. Air permeability classes

6.2 Watertightness

In conformity with § 4 of standard EN 12208:1999 and with reference to EN 1027:2016, the sample was classified with reference to the following table:

Test Pressure	Classification		
P_{max} in (Pa)	Test method A	Test method B	
-	**0**	**0**	No requirement
0	**1 A**	**1 B**	Sprinkling for 15 min
50	**2 A**	**2 B**	Same as class 1 + 5 min
100	**3 A**	**3 B**	Same as class 2 + 5 min
150	**4 A**	**4 B**	Same as class 3 + 5 min
200	**5 A**	**5 B**	Same as class 4 + 5 min
250	**6 A**	**6 B**	Same as class 5 + 5 min
300	**7 A**	**7 B**	Same as class 6 + 5 min
450	**8 A**	-	Same as class 7 + 5 min
600	**9 A**	-	Same as class 8 + 5 min
> 600	**Exxx**	-	Above 600 Pa with 150 Pa increments, each step shall last 5 minutes
Note: method A is fit for fully exposed products; method B is fit for partially protected products.			

Tab. 2. Watertightness classes

6.3 Resistance to wind load

In conformity with § 4, 5, 6 and 7 of standard EN 12210:2016 and with reference to EN 12211:2016, the sample was classified on the basis of the tables below (where P1, P2, P3 values are linked to each other by the following relations: P2 = 0.5 P1 and P3 = 1.5 P1). In order to globally classify the sample, the following requirements are to be first of all verified:

- no visible flaw shall be observed during a normal naked-eye examination at 1-meter distance, following both the first two tests (at P1 and P2 values);
- the sample shall maintain a satisfactory serviceability and the increase of air permeability shall result to be less than 20% as to the maximum air permeability admitted for the previously obtained class, following both the first two tests (at P1 and P2 values);
- the sample shall endure the safety test (at P3 value) without any disjunction or opening; the sample shall remain closed (glazing unit replacement and test repetition in case of a specific failure are however envisaged; likewise, the presence of defects such as bending and/or twist of accessories and cracking of parts of the frame are admitted).

Class	P1 (Pa)	P2 (Pa)	P3 (Pa)
0	Not subjected to test		
1	400	200	600
2	800	400	1200
3	1200	600	1800
4	1600	800	2400
5	2000	1000	3000
E_{xxxx}	xxxx		

Tab. 3. Wind load classes

Class	Relative frontal deflection
A	< 1/150
B	< 1/200
C	< 1/300

Tab. 4. Classes of relative frontal deflection

Wind pressure class	Relative frontal deflection		
	A	B	C
1	**A1**	**B1**	**C1**
2	**A2**	**B2**	**C2**
3	**A3**	**B3**	**C3**
4	**A4**	**B4**	**C4**
5	**A5**	**B5**	**C5**
E_{xxxx}	**AE_{xxxx}**	**BE_{xxxx}**	**CE_{xxxx}**

Tab. 5. Classes of resistance to wind load

7 Results

7.1 Preventive control on the sample (dimensions and surfaces)

Measures	width (m)	height (m)	surface (m^2)	opening joints length (m)
Whole sample	2,000	2,400	4,800	-
Opening part	1,909	2,309	4,408	10,745

Tab. 6

7.2 Air permeability test

Note: Uncertainty of measurement values are expressed as extended uncertainty (U) with coverage factor k=2, corresponding to a confidence level of 95%; classification is made according to UNI EN 12207:2017 without considering the uncertainty of measurement.

IRCCOS TECHNICIAN	DATE OF TEST	LABORATORY ENVIRONMENTAL PARAMETERS		
		Temperature (°C)	Relative humidity (%)	Atmospheric pressure (kPa)
Damiano Zizza	2020/11/27	T_x= 15,2	U_{rel}= 64,8	P_x= 99,5

Tab. 7

Pressure	Air permeability of the sample (positive pressure test)		
Pa	m^3/h	m^3/hm^2	m^3/hm
50	2,40	0,50	0,22
100	3,10	0,64	0,29
150	3,69	0,77	0,34
200	4,29	0,89	0,40
250	6,19	1,29	0,58
300	7,59	1,58	0,71
450	10,39	2,16	0,97
600	16,38	3,41	1,52

Tab. 8

Pressure	Air permeability of the sample (negative pressure test)		
Pa	m^3/h	m^3/hm^2	m^3/hm
50	3,10	0,64	0,29
100	3,10	0,64	0,29
150	3,10	0,64	0,29
200	3,99	0,83	0,37
250	4,59	0,96	0,43
300	5,69	1,19	0,53
450	8,89	1,85	0,83
600	12,78	2,66	1,19

Tab. 9

Pressure	Air permeability of the sample (arithmetic average of the two tests)		
Pa	m^3/h	m^3/hm^2	m^3/hm
50	2,75	0,57	0,26
100	3,10	0,64	0,29
150	3,40	0,71	0,32
200	4,14	0,86	0,39
250	5,39	1,12	0,50
300	6,64	1,38	0,62
450	9,64	2,01	0,90
600	14,58	3,04	1,36

Tab. 10

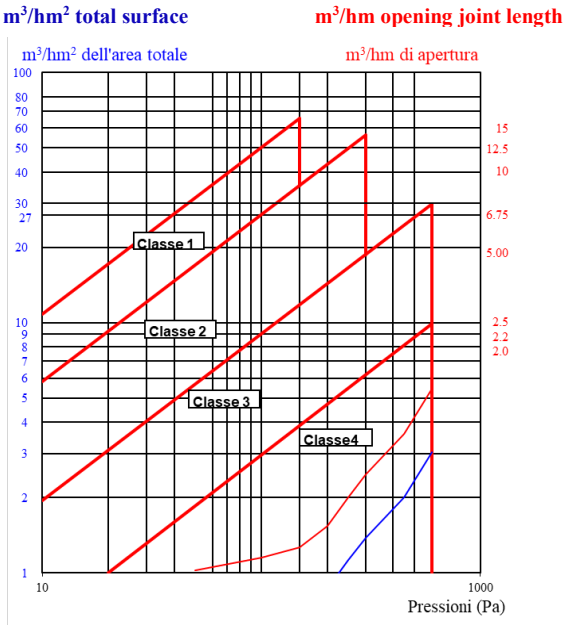


Diagramme 1

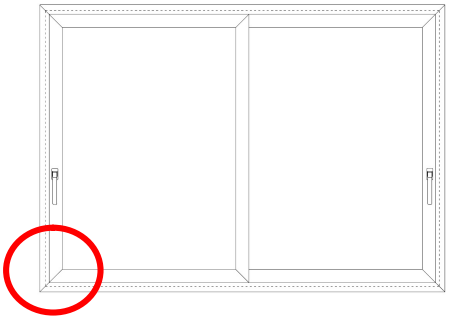
7.2.1 Classification of the sample

The sample subjected to air permeability test under positive and negative pressures was classified in class **4**.

7.3 Watertightness test

IRCCOS TECHNICIAN	DATE OF TEST	LABORATORY ENVIRONMENTAL PARAMETERS		
		Temperature (°C)	Relative humidity (%)	Water Temperature (°C)
Damiano Zizza	2020/11/27	T_x= 15,7	U_{rel}= 61,5	T_a = 16,3

Tab. 11

Pressure (Pa)	Duration (min)	Remarks
0	15	No water seepage
50	5	
100		
150		
200		
250		
300		
450		
600	1’	 

Tab. 12

7.3.1 Classification of the sample

The sample subjected to watertightness test was classified in class **8A**.

7.4 Resistance to wind load test

7.4.1 Deformation test (under positive and negative pressure)

IRCCOS TECHNICIAN	DATE OF TEST	LABORATORY ENVIRONMENTAL PARAMETERS		
		Temperature (°C)	Relative humidity (%)	Atmospheric pressure (kPa)
Damiano Zizza	2020/11/27	T_x= 18,8	U_{rel}= 48,0	P_x= 99,4

Tab. 13

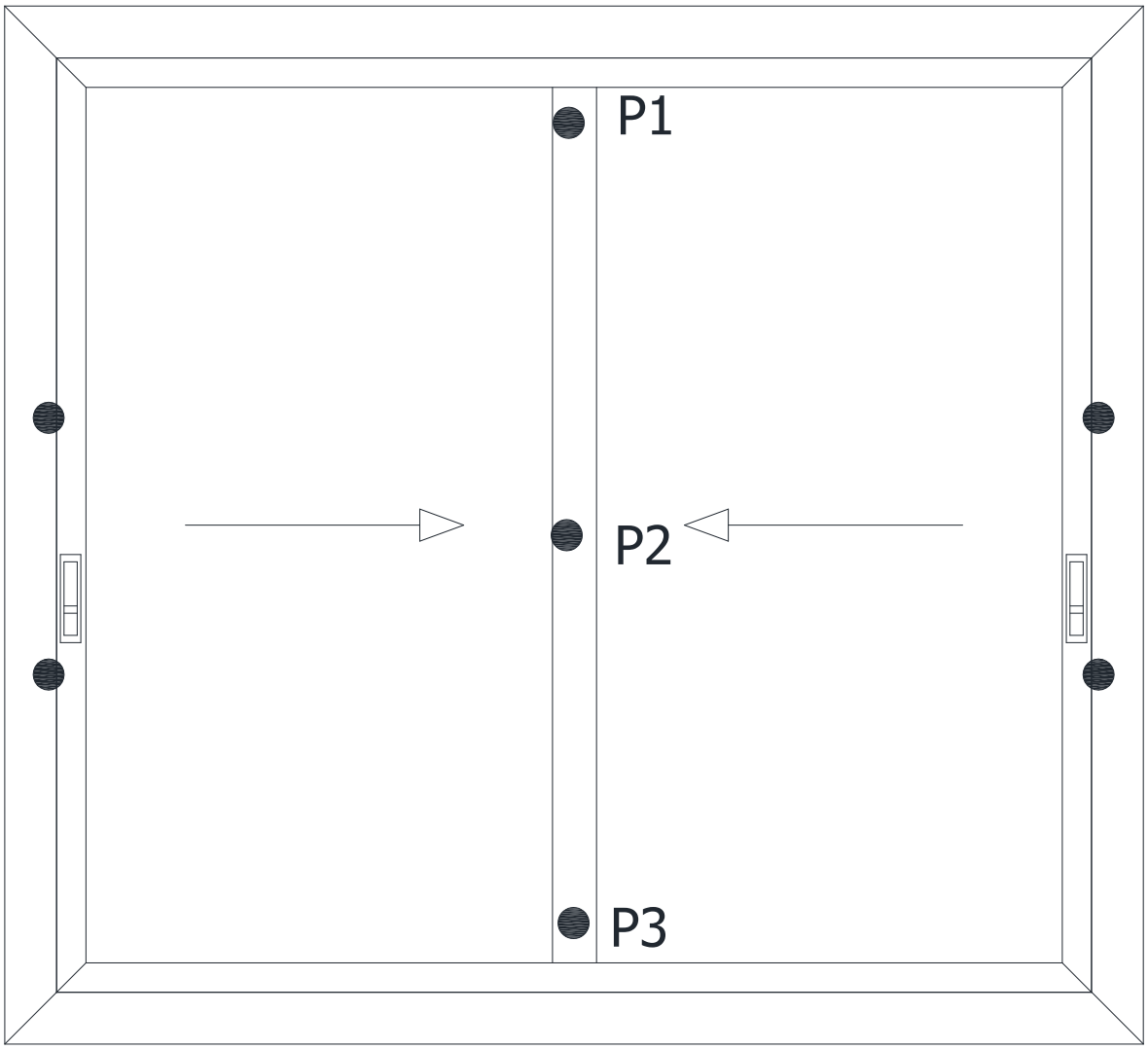


Fig. 1. Experimental setup for resistance to wind load test (inner view): transducers positioning scheme - placed on the inner side

dimensions (mm)	Mullion vertical
	2400

Tab. 13

Positive pressure (Pa)	**p. 1 (mm)**	**p. 2 (mm)**	**p. 3 (mm)**
1200	5,68	15,38	4,33
0	0,07	1,12	0,21
Negative pressure (Pa)	**p. 1 (mm)**	**p. 2 (mm)**	**p. 3 (mm)**
1200	7,20	18,16	6,00
0	1,44	5,28	2,17

Tab. 14. Frontal displacements of measured characteristic points at different test pressure steps

Mullion vertical	Positive pressure (Pa)	Frontal displacements (mm)			Frontal deflection (mm)	Relative frontal deflection
		p.3(left)	p.4 (center)	p.5 (right)		
	1200	5,68	15,38	4,33	10,38	**1/231**
		Residual Deformation (mm)			Frontal residual def. (mm)	
	0	0,07	1,12	0,21	0,98	
	Negative pressure (Pa)	Frontal displacements (mm)			Frontal deflection (mm)	Relative frontal deflection
		p.3(left)	p.4 (centr)	p.5 (right)		
	1200	7,20	18,16	6,00	11,56	**1/208**
		Residual Deformation (mm)			Frontal residual def. (mm)	
	0	1,44	5,28	2,17	3,48	

Tab. 15. Frontal deflections and residual deformations of left mullion of the sample under test

7.4.1.1 Remarks on results

At the end of the deformation test no visible defect was observed during a normal naked-eye examination at 1-meter distance and the sample maintained a satisfactory serviceability. The relative frontal deflection of the most deformed element of the tested sample resulted to be < **1/200** (cf. Tab. 4).

7.4.2 Repeated pressure test

The sample was subjected to 50 cycles including negative and positive pressures at ± 600 Pa.

7.4.2.1 Remarks on results

At the end of the deformation test no visible defect was observed during a normal naked-eye examination at 1-meter distance and the sample maintained a satisfactory serviceability.

7.4.3 Verification of air permeability

Note: Uncertainty of measurement values are expressed as extended uncertainty (U) with coverage factor k=2, corresponding to a confidence level of 95%; classification is made according to UNI EN 12207:2017 without considering the uncertainty of measurement.

IRCCOS TECHNICIAN	DATE OF TEST	LABORATORY ENVIRONMENTAL PARAMETERS		
		Temperature (°C)	Relative humidity (%)	Atmospheric pressure (kPa)
Damiano Zizza	2020/11/27	T_x= 18,7	U_{rel}= 48,2	P_x= 99,3

Tab. 16

Pressure	Air permeability of the sample (positive pressure test)		
Pa	m^3/h	m^3/hm^2	m^3/hm
50	2,86	0,59	0,27
100	3,74	0,78	0,35
150	4,73	0,98	0,44
200	6,01	1,25	0,56
250	7,88	1,64	0,73
300	9,26	1,93	0,86
450	17,13	3,57	1,59
600	21,86	4,55	2,03

Tab. 17

Pressure	Air permeability of the sample (negative pressure test)		
Pa	m^3/h	m^3/hm^2	m^3/hm
50	3,45	0,72	0,32
100	3,25	0,68	0,30
150	4,92	1,03	0,46
200	5,12	1,07	0,48
250	6,99	1,46	0,65
300	9,65	2,01	0,90
450	16,15	3,36	1,50
600	21,96	4,57	2,04

Tab. 18

Pressure	Air permeability of the sample (arithmetic average of the two tests)		
Pa	m^3/h	m^3/hm^2	m^3/hm
50	3,15	0,66	0,29
100	3,50	0,73	0,33
150	4,82	1,01	0,45
200	5,56	1,16	0,52
250	7,43	1,55	0,69
300	9,45	1,97	0,88
450	16,64	3,47	1,55
600	21,91	4,56	2,04

Tab. 19

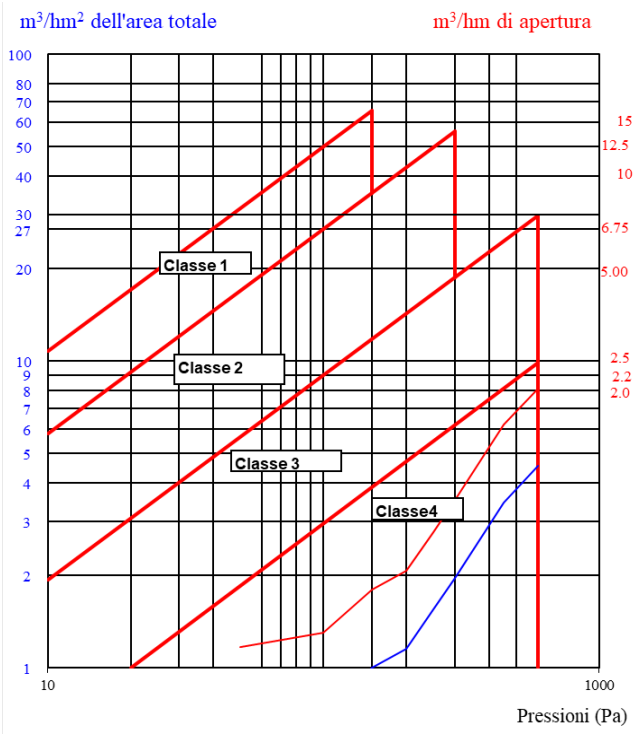


Diagramme 2

7.4.3.1 Remarks on results

The requirement of the containment of the maximum increase of the air permeability observed within 20% in comparison to the maximum allowed air permeability for the class previously obtained was complied with.

7.4.4 Safety test

IRCCOS TECHNICIAN	DATE OF TEST	LABORATORY ENVIRONMENTAL PARAMETERS		
		Temperature (°C)	Relative humidity (%)	Atmospheric pressure (kPa)
Damiano Zizza	2020/11/27	T_x= 19,1	U_{rel}= 49,0	P_x= 99,1

Tab. 20

	Observed structural damages or degradations
n° 1 gust at + 1800 Pa	none
n° 1 gust at – 1800 Pa	none

Tab. 21

7.4.4.1 Remarks on results

At the end of the safety test, no disjunction or functional degradation of the sample was observed. The sample remained closed

7.4.5 Classification of the sample

The sample subjected to resistance to wind load test was classified in class **B3.**

8 Photographs of the sample under test and of the experimental setup



Photo 1: Tested sample in the experimental setup



Photo 2: Tested sample in the experimental setup undergoing wind load resistance tests

9 Resume of classes assigned on the basis of the results of the tested sample of a Double-casement sliding door with trade name "PB2AAR_EVEREST_91000", according to TR No 1994-CPR-RP2157, issued on 2020/01/13 to EVEREST SHPK .

Test	Test standard	Classification standard	Obtained Class
Air permeability	EN1026:2016	EN 12207:2017	class **4**
Watertightness	EN1027:2016	EN 12208:1999	class **8A**
Resistance to wind load	EN12211:2016	EN 12210:2016	class **B3**

10 Retention of samples

The test sample has to be retained at least until the Test Report emission.

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---------- End of the Test Report No. 1994-CPR-RP2157----------